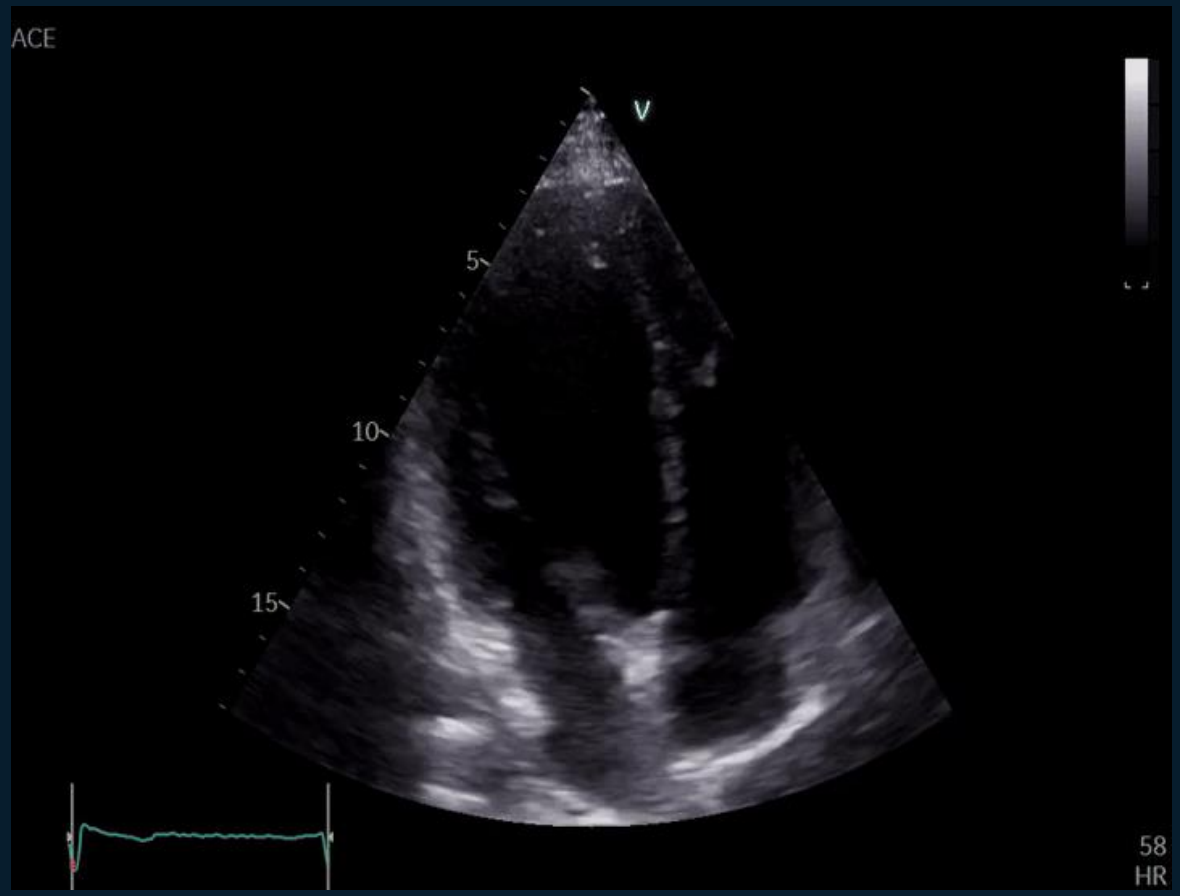
HOW DOES A RESEARCH PROJECT EVOLVE?

From a clinical question to a research program



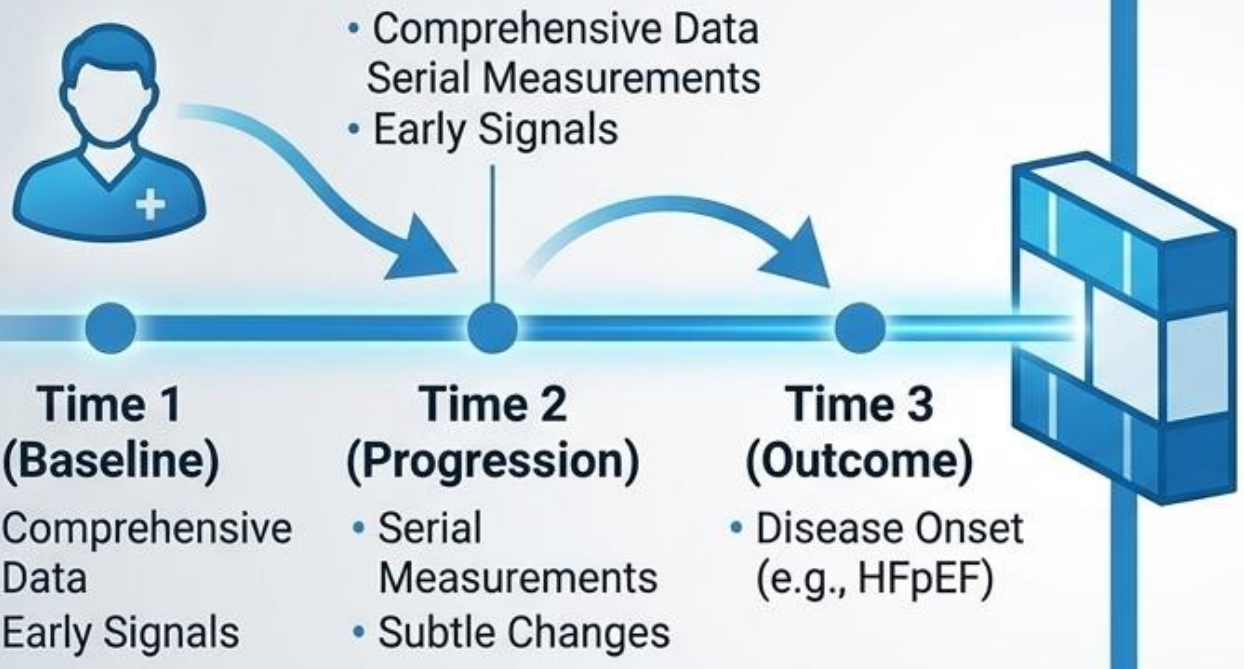
A patient looked healthy. Months later, the story changed.

FIRST VISIT	MONTHS LATER	THE QUESTION
Chest pain Routine exams Reassuring Discharged	Returned with more serious heart disease, specifically HFpEF	Was something subtle already present in the earlier echo?

Could machine learning detect an early signal that was missed?

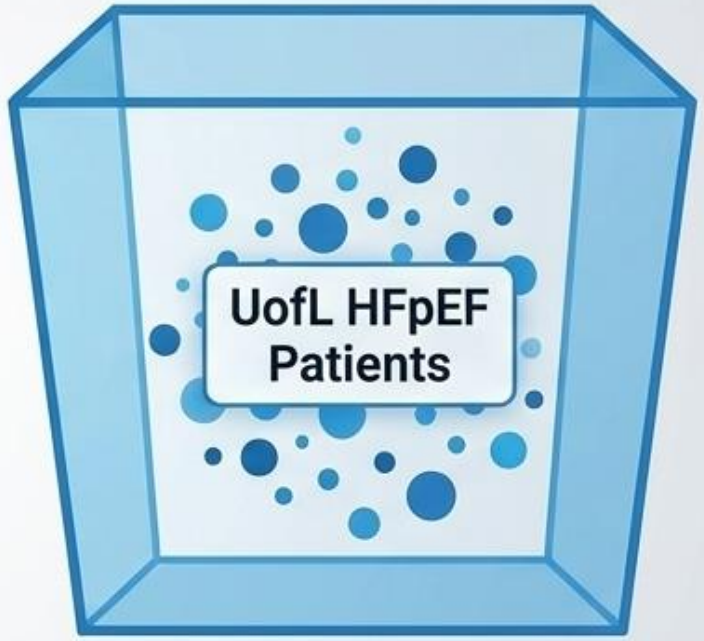
The Ideal Data

Longitudinal Cohort



Our Reality

Cross-sectional Snapshot

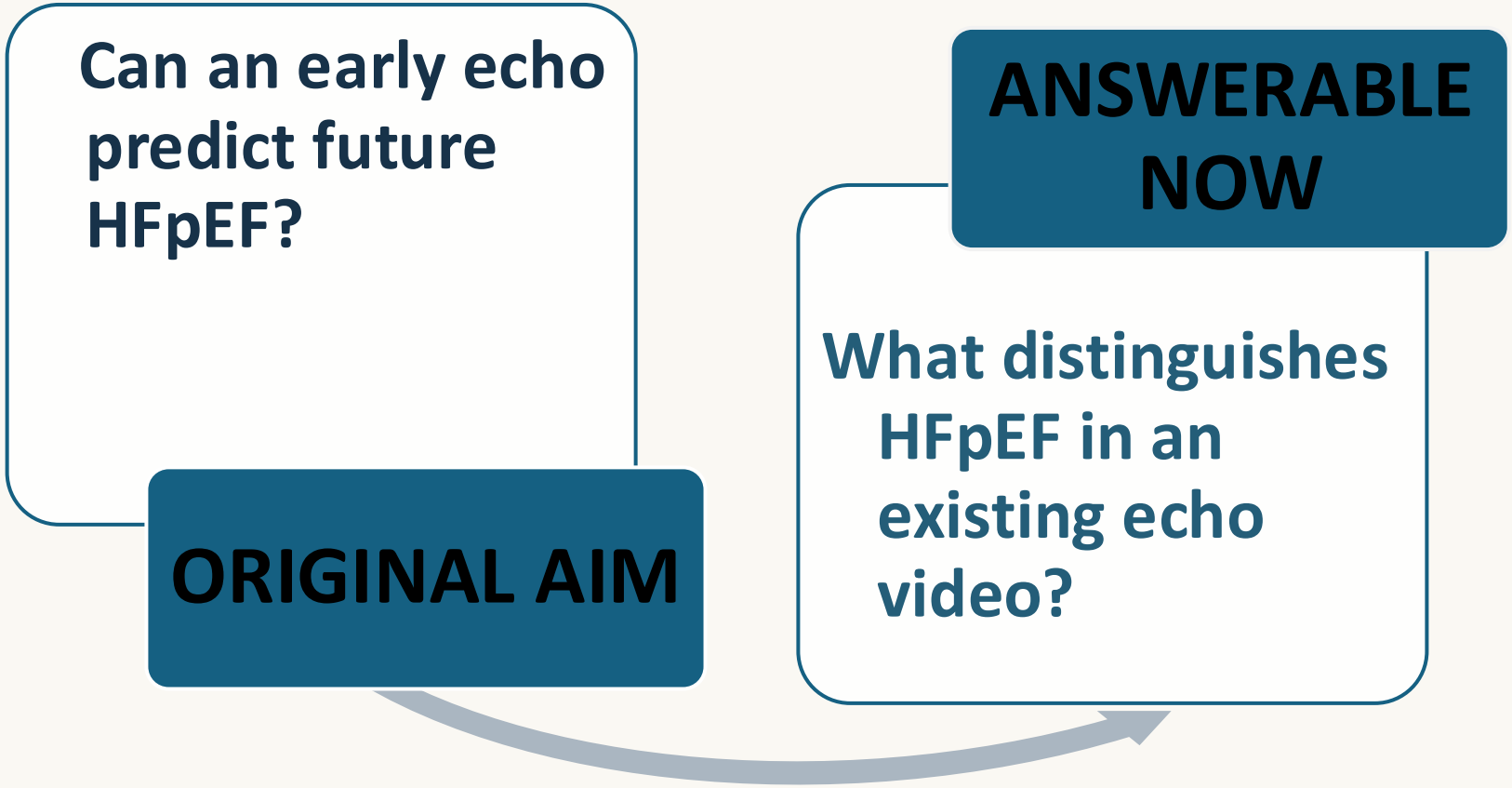


We could not train a model for chronological early prediction without chronological data. The initial path was blocked.

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THE PIVOT

We changed the question to one we could test



Our HFpEF group had no local controls yet

UofL

≈200 HFpEF studies

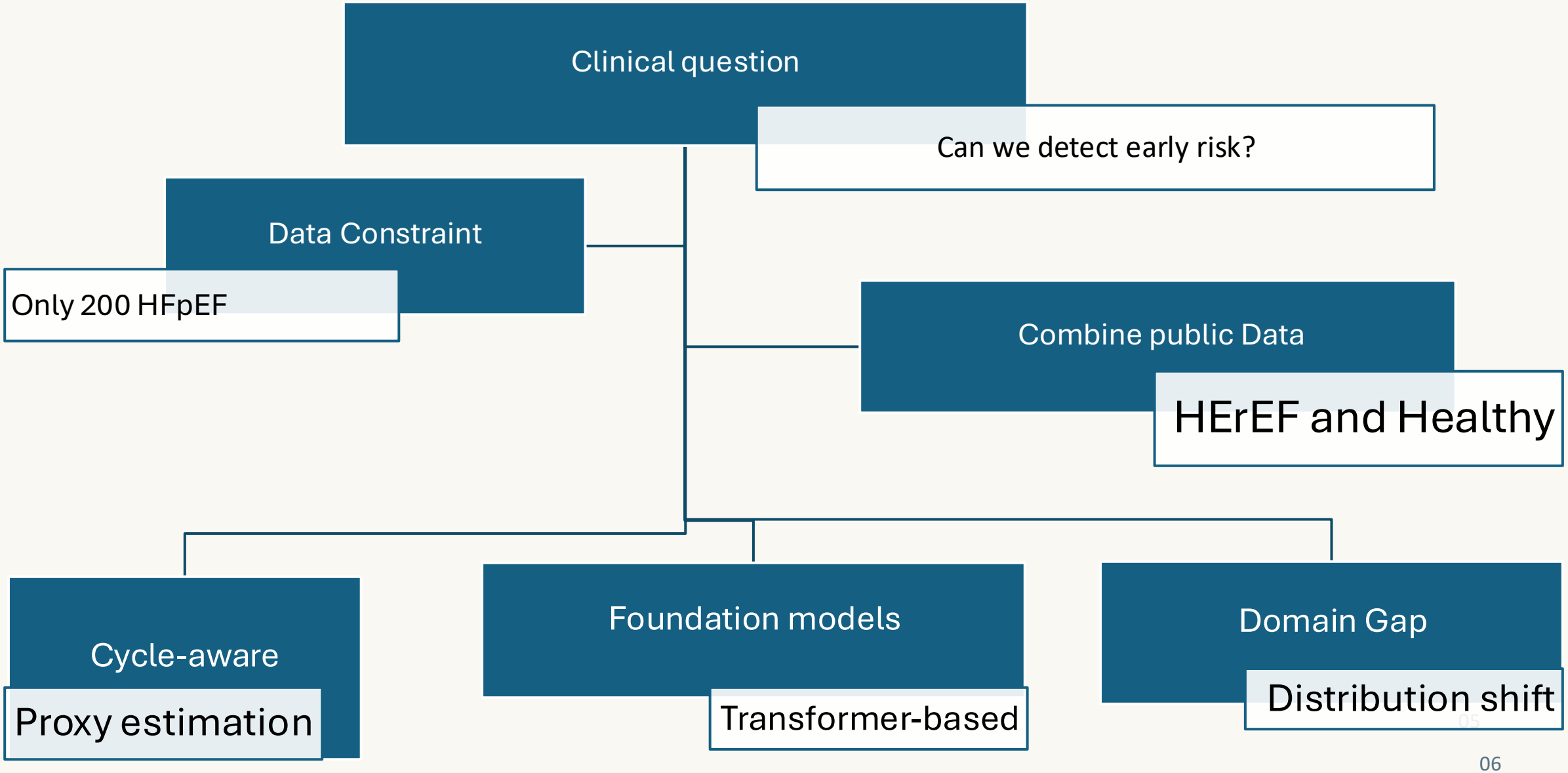
PUBLIC DATA

Healthy + HFrEF

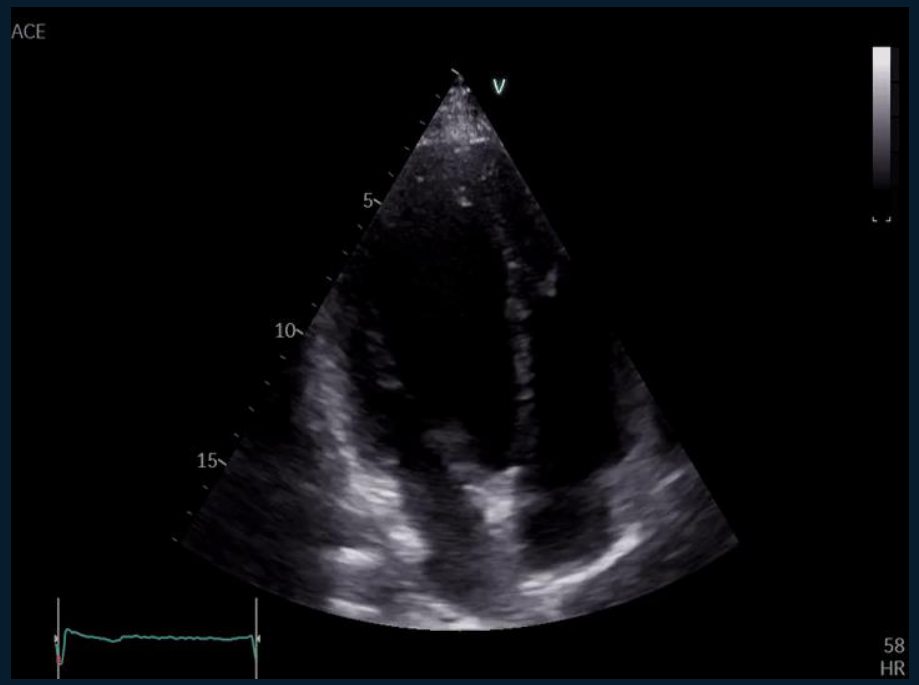
Disease label and data source were entangled.

A model might learn hospital or acquisition differences instead of pathology.

Each answer created the next question



Could the model learn from the moving heart?



Echo video

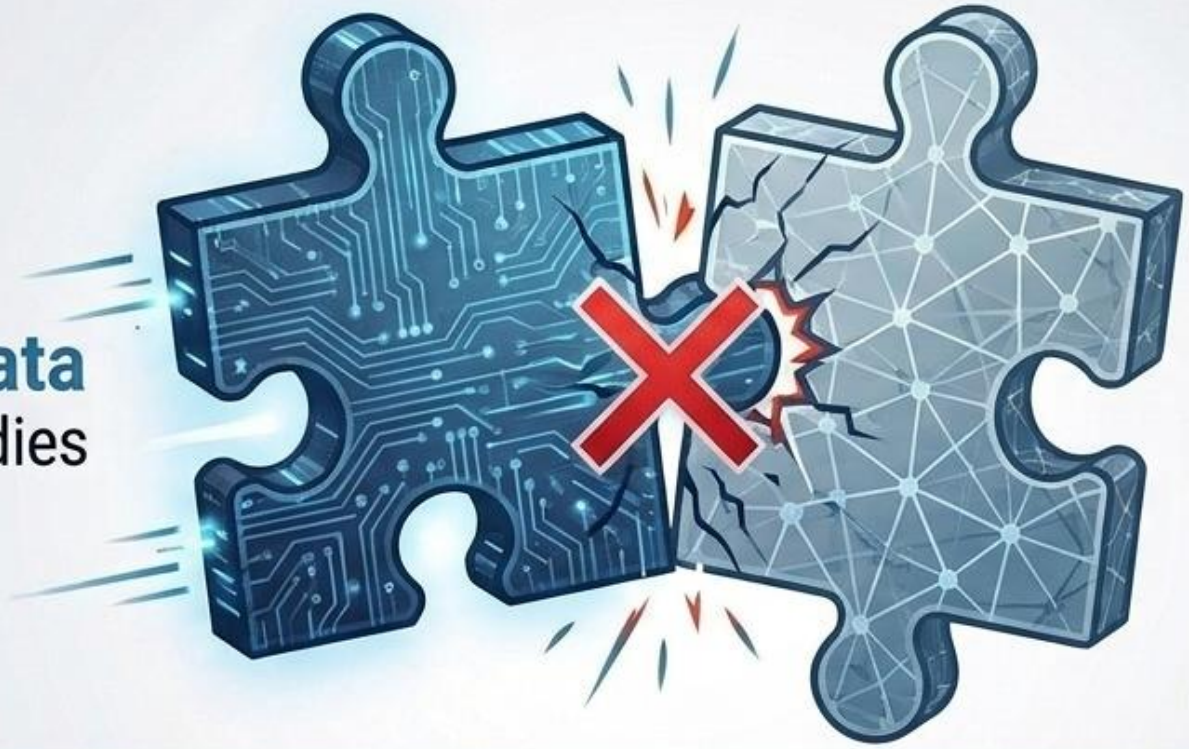
Estimate cycles and ED / ES

TimeSformer video model

The ED and ES frames came from maximum/minimum image proxies,
not manual annotations.

The Comparison Problem

Local UofL Data
≈200 HFpEF studies

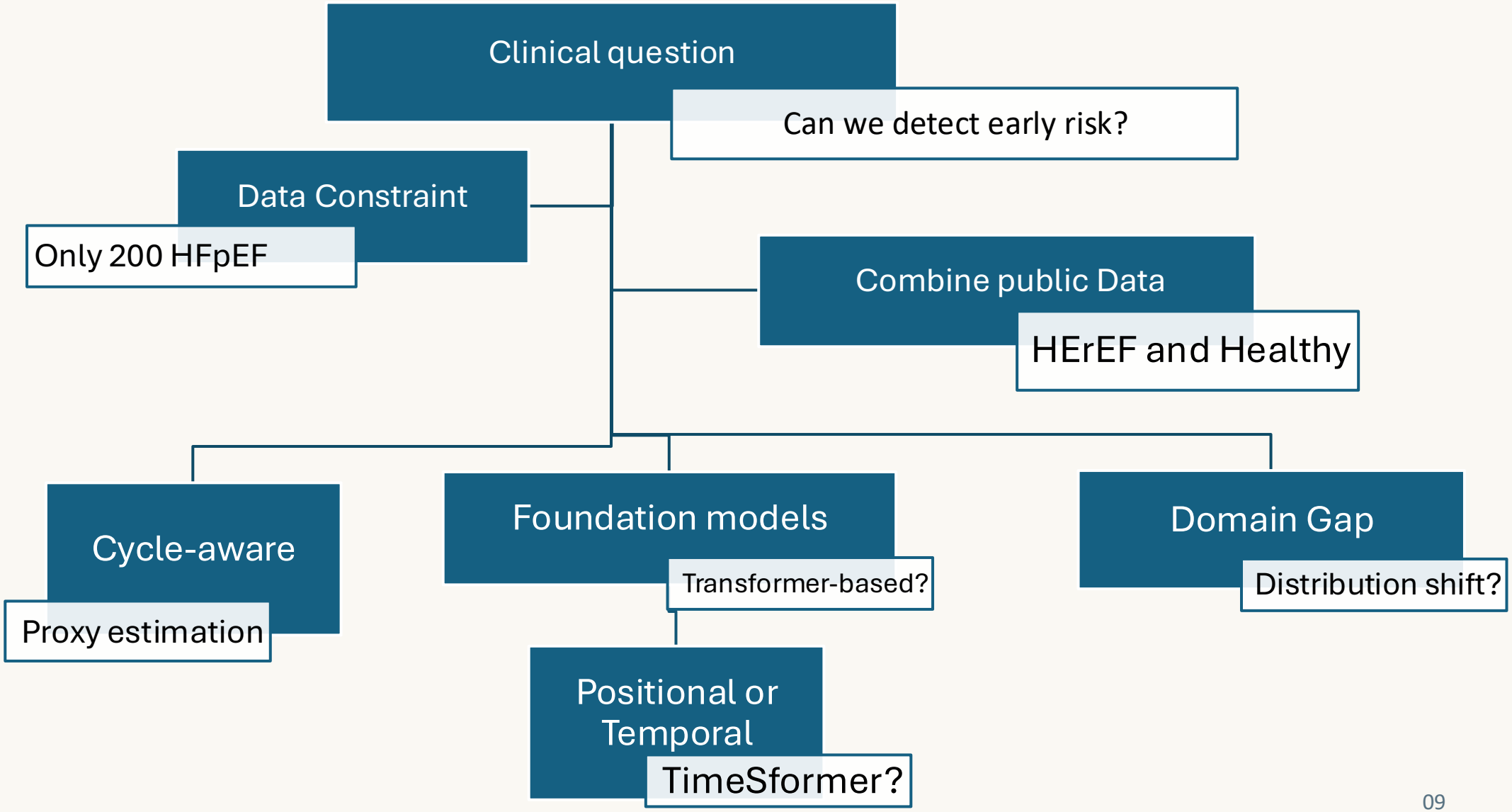


Public Data Cohort
Healthy + HFrEF

The Comparison Problem: The disease labels became perfectly entangled with the data sources. We risked building a model that learns hospital acquisition differences instead of actual cardiac pathology.

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How can fine-tuning a foundation model improve diagnosis?



Public Cohort Only

Healthy vs. HFrEF (619 studies)

Action: Stripped away the local HFpEF data to eliminate domain shift variables.

EMBC 2026 TimeSformer study published.

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New local data unlocked a cleaner three-class study

HEALTHY	HFrEF	HFpEF
≈200	≈200	≈200

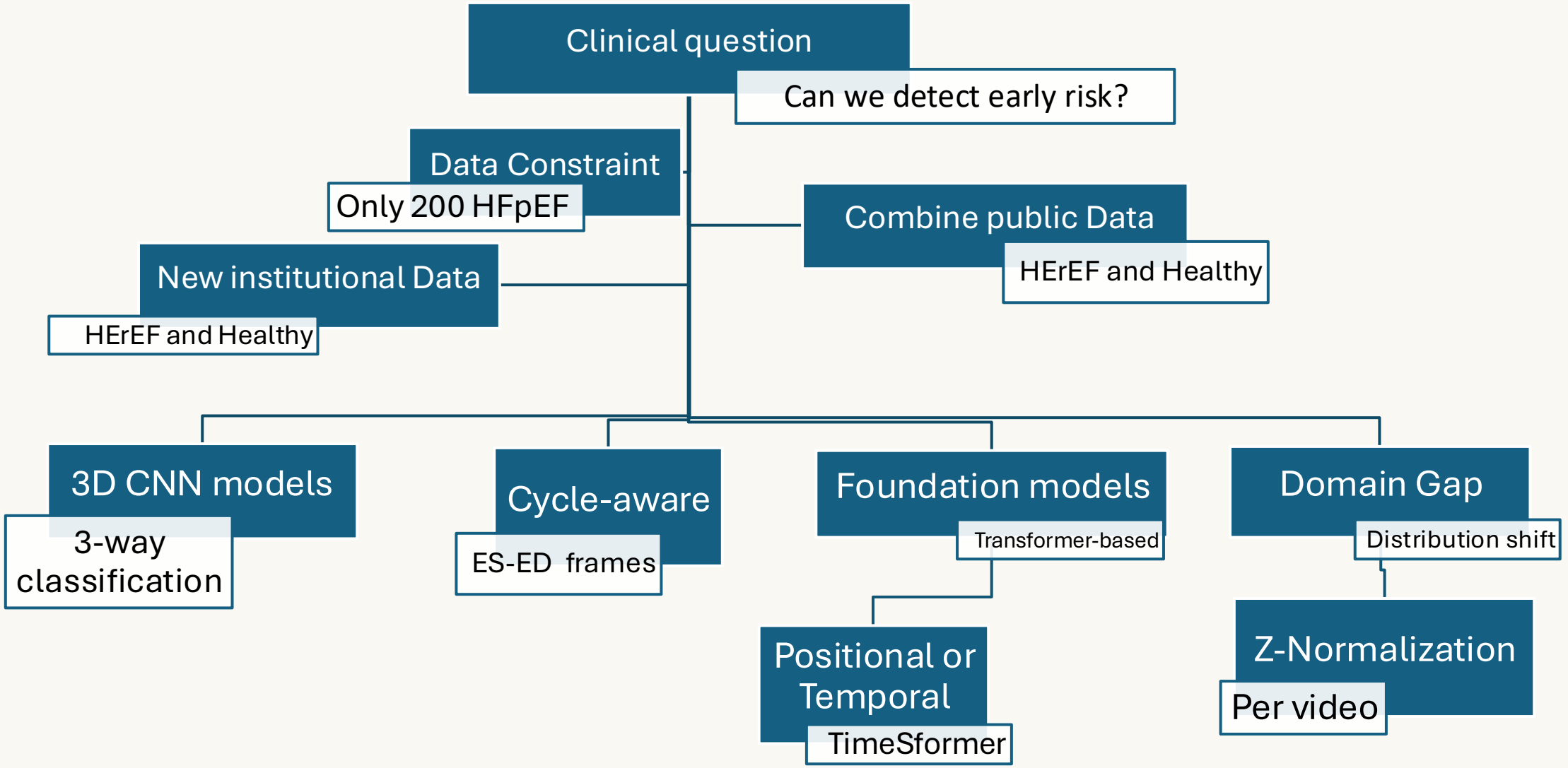
We trained 3D CNN models locally, then tested on a public cohort.



The Discovery: Testing our local model on public data revealed a harsh distribution shift. The model struggled to adapt to diverse outside demographics and varied scanning protocols.

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Each answer created the next question



Public Cohort Only

EMBC 2026 TimeSformer study published.



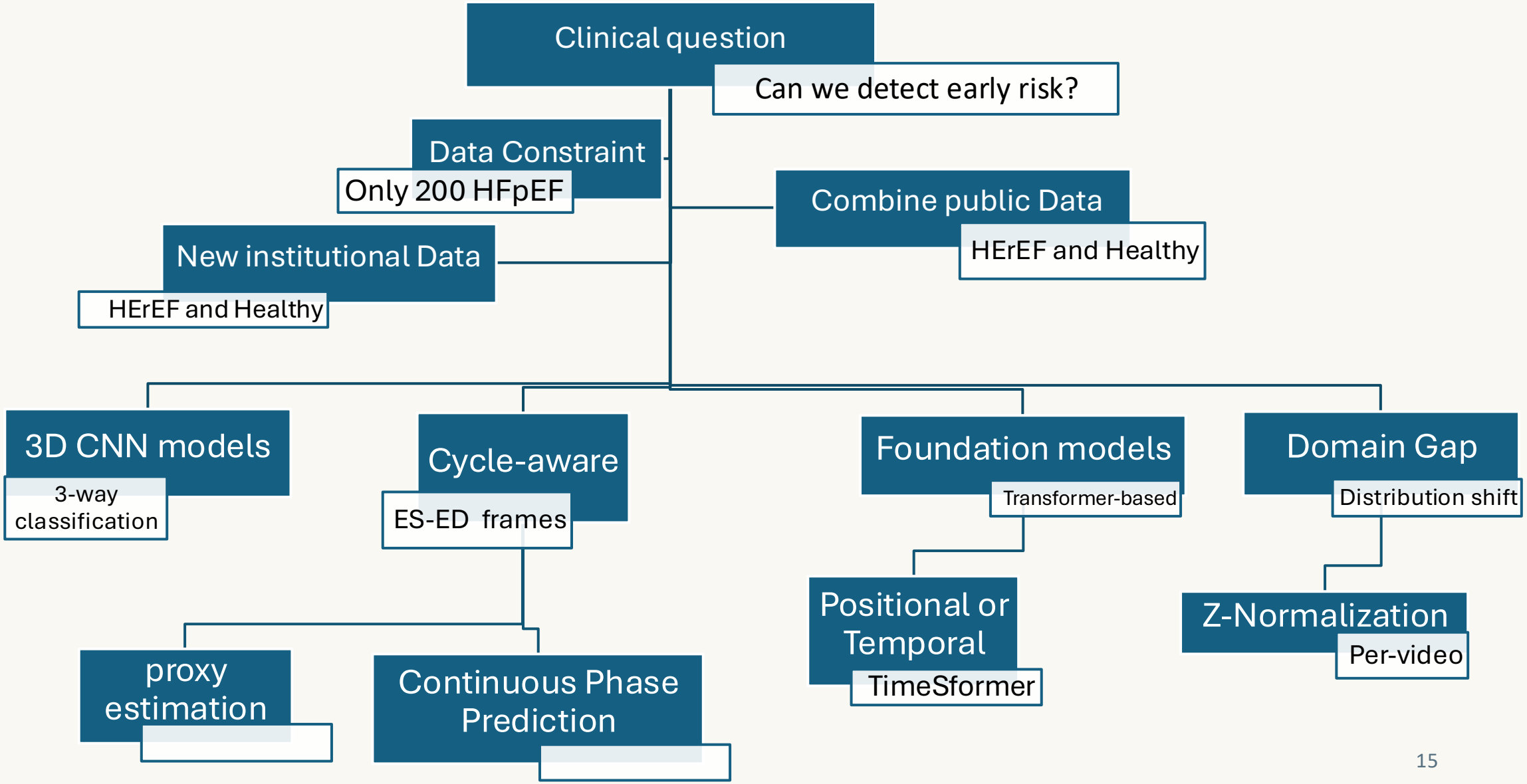
All perfectly matched local UofL data.

Trained **3D CNN** models locally from scratch.

Achieved a clean three-class study without source entanglement. **Second paper published.**

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Can we trust proxy-estimation method?



THE NEW FINDING

External testing changed the research question again

TRAINED HERE

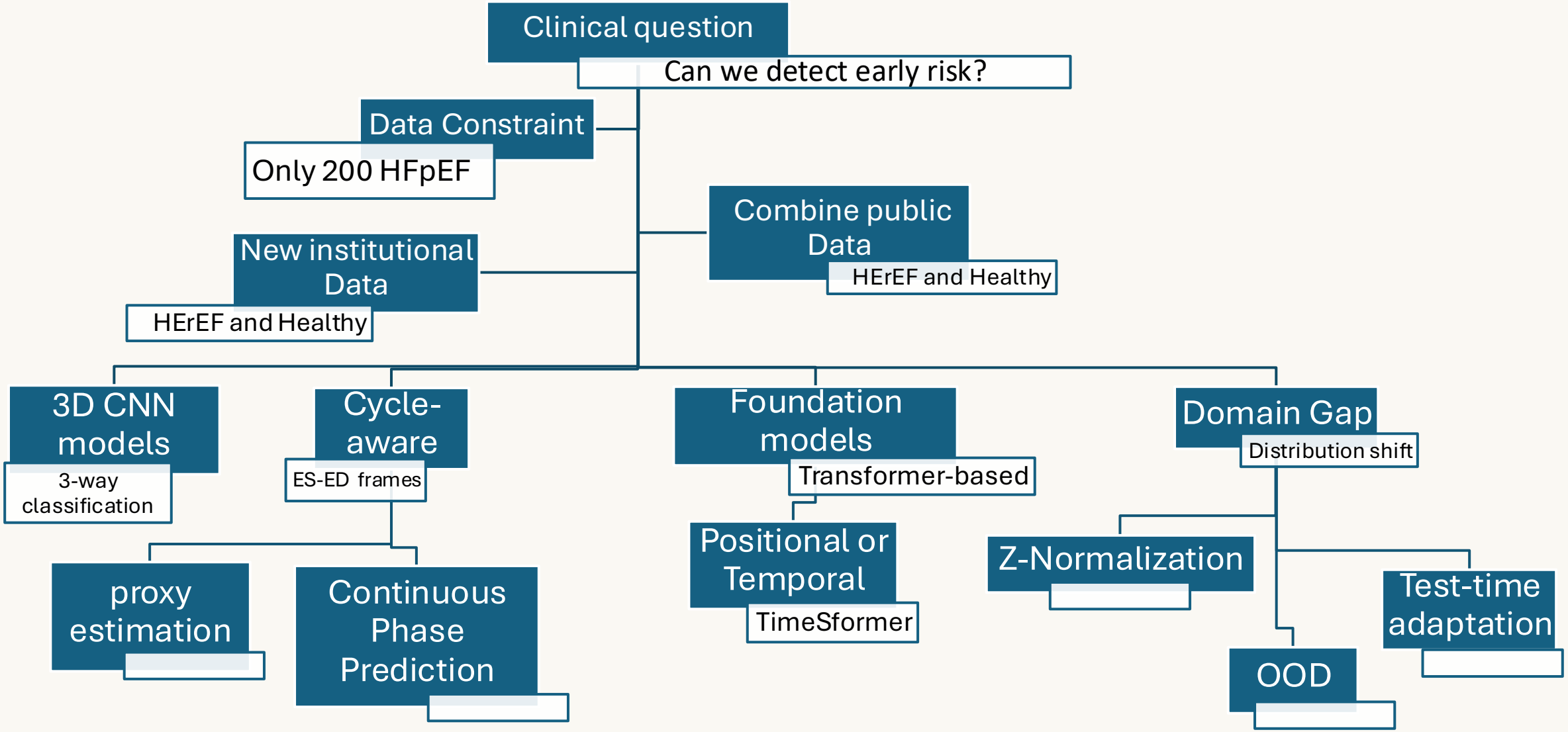
UofL cohort

TESTED THERE

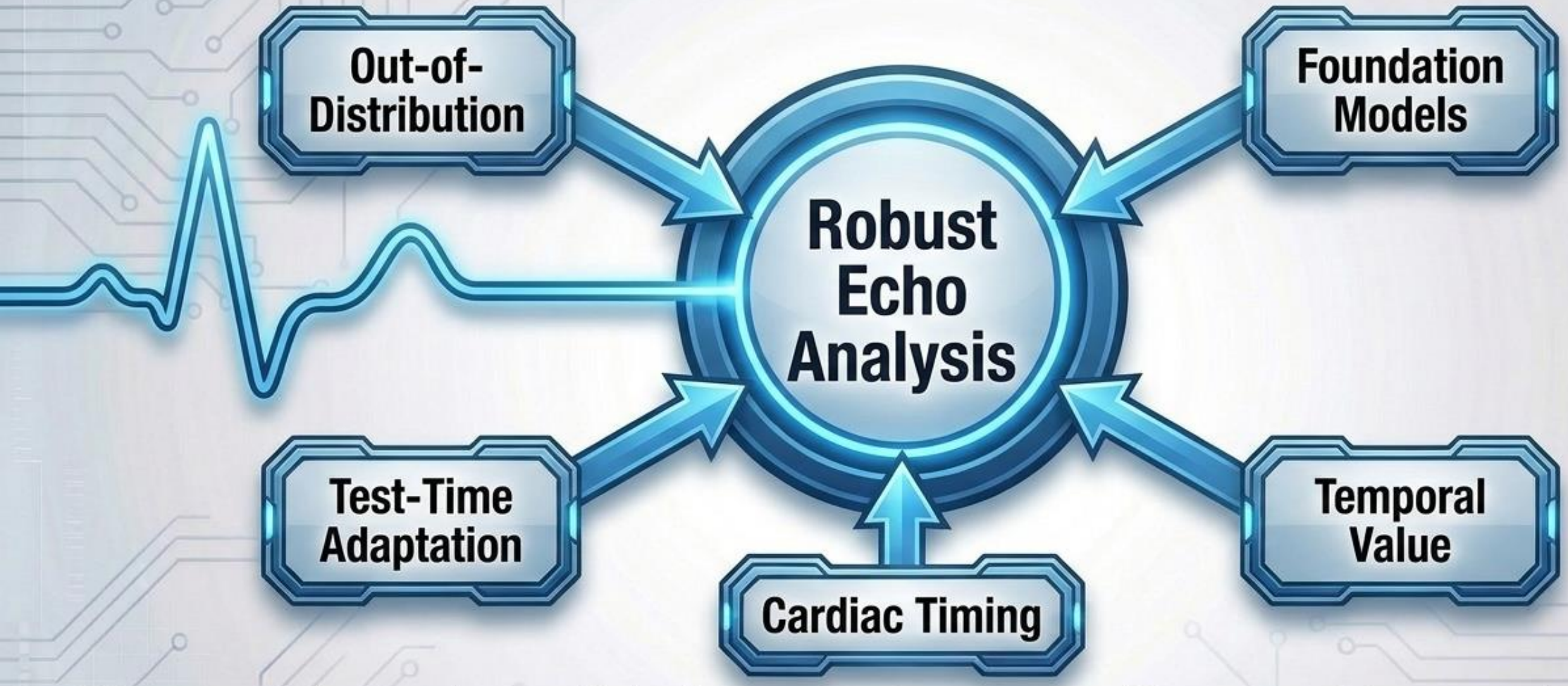
Public cohort

Why did performance change across datasets?

How to generalize our model and overcome domain gaps



The Research Program Architecture



What started as a single clinical question has matured into a multi-pronged research program addressing the fundamental challenges of medical deep learning.

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